

An adversarial discriminative temporal convolutional network for EEG-based cross-domain emotion recognition

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ABSTRACT

Domain adaptation (DA) tackles the problem where data from the source domain and target domain have different underlying distributions. In cross-domain (cross-subject or cross-dataset) emotion recognition based on EEG signals, traditional classification methods lack domain adaptation capabilities and have low performance. To address this problem, we proposed a novel domain adaptation strategy called adversarial discriminative-temporal convolutional networks (AD-TCNs) in this study, which can ensure the invariance of the representation of feature graphs in different domains and fill in the differences between different domains. For EEG data with specific temporal attributes, the temporal model TCN is used as the feature encoder. In the cross-subject experiment, our AD-TCN method achieved the highest accuracies of the valence and arousal dimensions in both the DREAMER and DEAP datasets. In the cross-dataset experiment, two of the eight task groups showed accuracies of 62.65% and 62.36%. Compared with the state-of-the-art performance in the same protocol, experimental results demonstrated that our method is an effective extension to realize EEG-based cross-domain emotion recognition.

1. Introduction

Emotion plays an important role in human-computer interactions. Emotion recognition is an indispensable part of humanization, involving every aspect of daily life. Emotional state is the comprehensive expression of a person's behavior and psychology in a certain environment. We can identify emotional states in terms of behavior, expression, language, and so on. In our previous work [1], a comparative analysis for multi-modal emotion recognition based on brain-computer interfaces was performed. We found that electroencephalograms (EEGs) can reflect emotional states more objectively than behavioral modalities. Many studies [2,3] have also shown that EEGs can reflect emotional states.

However, because EEG signals have the characteristics of individual differences and are nonstationary due to the inherent diversity of mental states, the degree of response to stimuli among different subjects, the shape of the head and skull, etc. [4–6], it is difficult to train a generic model that can be applicable to different datasets or subjects [7]. Therefore, training a specific model for each subject is the most common approach to recognize emotion based on EEG [8,9]. However, to reach satisfactory recognition performance, researchers need to collect sufficient labeled data (generally more than 1 h) before specific model

training [10–12]. The laborious and time-consuming labeling procedure for the new subject has become a major bottleneck for practical uses of EEG-based emotion recognition [12].

Considering the foregoing shortcoming, it is very important to realize cross-subject emotion recognition based on EEG signals in human-computer interaction applications. We refer to the EEG of different subjects or datasets as individual domains. Due to the individual differences and nonstationarity of EEG signals, the EEGs of different domains obey different distributions, which is referred to as a domain shift [5,12–14]. Traditional machine learning algorithms depend on the assumption that the training and test data are independent and identically distributed [15,16]. Therefore, the performance of the trained traditional classifier will often decline sharply for the new subject or new dataset.

To address the above issues, one promising method is domain adaptation (DA) [12], which aims to eliminate the differences in the distribution of EEG data between the source domain and target domain and further learn the extracted transferable features for emotion recognition. Fig. 1 shows an example: in an EEG-based emotion recognition system, if the training samples and test samples come from different data domains (x_s , x_t), the classifiers (c_s) trained by the source

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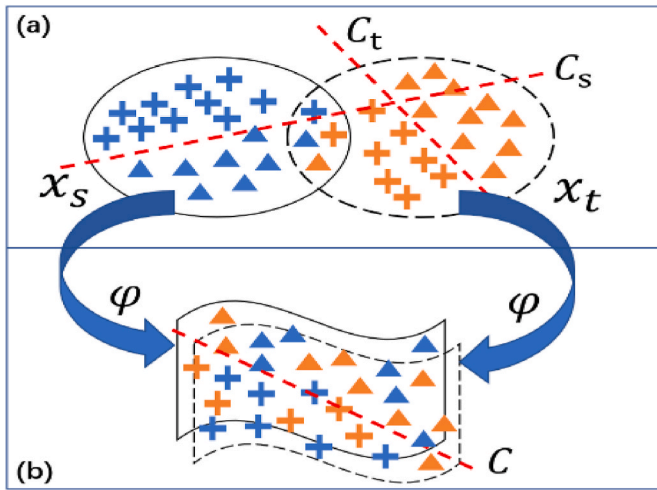


Fig. 1. (a) Source domain feature data distribution x_s , target domain feature data distribution x_t ; (b) Domain adaptation ϕ to matched distributions of the source domain and target domain.

domain are not adapted to the target domain. After adaptation (ϕ), both the source domain and target domain can effectively complete the task of EEG-based emotion recognition under the same classifier (c). DA could improve learning in the unlabeled target domain through the transfer of knowledge from the source domains [17].

In this paper, we propose a new model called adversarial discriminative temporal convolutional networks (AD-TCNs) for EEG-based cross-domain emotion recognition, which aims to eliminate the complicated steps of target domain labeling and improve the performance by unlabeled target domain data. For EEG signals with specific temporal attributes, we used the temporal model TCN as the feature encoder. The AD-TCN is used to learn an asymmetric mapping that adapts the target domain feature encoder to the source domain, where the EEG data of the target domain and source domain are from different subjects or different datasets.

In summary, the main contributions of our research are presented as follows:

- 1) Adversarial adaptive learning based on asymmetric mapping is integrated into a temporal convolutional neural network simultaneously, which can accordingly benefit the emotion recognition performance of EEG-based cross-domains and significantly enhance distribution matching.
- 2) Compared with the performance of existing methods in the same protocol, experimental results demonstrated that our method is an effective extension to realize EEG-based cross-domain emotion recognition.

The remainder of this paper is organized as follows: Related studies on cross-domain emotion recognition are reviewed in Section 2. In Section 3, the proposed AD-TCN method is presented in detail. In Section 4, the emotional dimension, datasets, and data processing for EEG signals are described. In Section 5, two emotion recognition experiments are conducted, and the corresponding results are reported. Finally, the discussion and conclusions of this paper are given in Sections 6 and 7.

2. Related work

In this section, we briefly review the following three strategies to solve cross-domain problems: feature selection for cross-domain, conventional domain adaptation methods, and deep adversarial networks for cross-domain. All technical points in these reviewed studies and our work are summarized in Table 1.

2.1. Feature selection for cross-domain

Many cross-domain studies, from the perspective of feature selection, select effective new feature subsets from high-dimensional EEG features to search for features with strong commonality among individuals to improve the generalization of the cross-domain emotion recognition model. Jiang et al. [18] adopted a decision tree classifier based on sequential backward selection (DT-SBS). In an N-dimensional feature set, one feature was eliminated each time. Then, the decision tree classifier for classification obtains N recognition performances. Li et al. [19] automatically extracted 18 linear and nonlinear EEG features and obtained the highest two-class emotion recognition mean accuracy of 59.06% on the DEAP dataset. In Ref. [20], a subject-independent emotion recognition system based on variational mode decomposition (VMD) was proposed. They used VMD as a feature extraction technique,

Table 1
Technical points of reviewed literatures and our work.

Method Style	Studies	Model	Findings of technical point
Feature selection for cross-domain	Jiang et al. [18]	Decision tree classifier based on sequential backward selection (DT-SBS)	The cross-domain method based on feature selection is not only complicated in the steps of repeated evaluation but also limited to the case that the source domain and the target domain have a certain similar distribution.
	Li et al. [19]	SVM based on recursive feature elimination (SVM-RFE)	
Conventional domain adaptation methods	P. Pandey et al. [20]	MLP based on variational mode decomposition (MLP-VMD)	Since the representation ability of conventional domain adaptation methods is limited, the domain shift problem cannot be effectively solved when the data distribution is quite different.
	Pan et al. [21]	Transfer component analysis (TCA)	
	K. Müller [22]	Kernel principal component analysis (KPCA)	
	R. Collobert et al. [23]	Transductive SVM (T-SVM)	
Deep adversarial network	E. Sangineto et al. [24]	Transductive Parameter Transfer (TPT)	The source and target domains share a feature encoder to achieve a symmetric mapping of the features. Because the data of both the source domain and the target domain are entered into the network for feature extraction, the number of parameters is so small that easily lead to underfitting problems.
	Lew et al. [13]	Regionally operated domain adversarial network (RODAN)	
	Rayatdoost et al. [14]	Subject-Invariant domain and subject adaption EEG-based emotion recognition (SIDA-EEG)	
	Bao et al. [31]	Two-Level domain adaptation neural network (TDANN)	
	Our work	Adversarial discriminative temporal convolutional network (AD-TCN)	
			Adversarial adaptive learning based on asymmetric mapping is integrated into a temporal convolutional neural network, which utilize the temporal sequence modeling ability of TCN and eliminating interdomain difference's ability of deep adversarial network based on based asymmetric mapping.

and a deep neural network was used as the classifier.

The hypothesis of the effectiveness of the feature selection method is to assume that there is little difference between some feature samples in the source domain and the target domain. By repeatedly evaluating the difference between the current selected feature data distribution and the target domain data distribution, feedback is carried out according to the evaluation results, and the subsequent selection process is continuously instructed. Finally, a group of the most robust EEG features are determined to achieve cross-domain emotion recognition.

The cross-domain method based on feature selection is not only complicated in the steps of repeated evaluation but also limited to the case that the source domain and the target domain have a certain similar distribution.

2.2. Conventional domain adaptation methods

In previous studies, several traditional DA methods were applied in the BCI field. Zheng [12] compared the performance of four different conventional domain adaptation approaches (transfer component analysis (TCA) [21], kernel principal component analysis (KPCA) [22], transductive support vector machines (TSVMs) [23], and transductive parameter transfer (TPT) [24]) on the EEG-based emotional SEED dataset [3].

For TCA, it was proposed by Pan et al. [21]. It is an adaptive method based on marginal distribution that mainly solves the problem of probability distribution in transfer learning. The KPCA was first proposed by Schölkopf [22]. It is a nonlinear dimensionality reduction method that maps high-dimensional space to low-dimensional space by the kernel method. TSVMs are used to improve the generalization accuracy of support vector machines (SVMs) by using unlabeled data. It uses labeled training data to learn a large interval hyperplane classifier while forcing the hyperplane away from unlabeled data. The TPT is a kind of parameter-based domain adaptation method. It mainly constructs the regression model of the two through the relationship between the classifier and the data distribution. The classifier of the target domain is obtained using the regression model.

These methods are all learned in shallow space, and the depth information of the feature is not extracted, so the generalization ability is limited. Few studies have taken advantage of deep networks to solve the domain shift problem in the EEG-based emotion recognition field.

2.3. Deep adversarial network for cross-domain

For deep adversarial networks, generative adversarial networks (GAN) was first proposed by Goodfellow et al. [25], which combined deep networks and adversarial learning to generate similar images. The GAN simultaneously trains a generative network G and a discriminative network D . Generally, the training process of the adversarial generation model allows the generator to "deceive" the discriminator as much as

possible, and the discriminator recognizes the forged samples generated by the generator as much as possible. There is competition between the generator and discriminator. The competitive discriminator tries its best to identify the forged samples generated by the generator. Ideally, when the training is over, the generator has fully learned the distribution of real samples and ensures that the model cannot distinguish between true and false data distributions.

Inspired by GANs, in the field of computer vision and natural language processing, many methods [26–30] of deep domain adaptation have been proposed to solve the problem of domain shift. According to the relationship between the source domain feature encoder and target domain feature encoder, the methods of deep adversarial domain adaptation can be divided into two ways: symmetric mapping [28] and asymmetric mapping [29]. For the current studies of EEG-based cross-domain emotion recognition based on deep adversarial learning, symmetric mapping is mainly used [11,13,31]. Lew et al. [13] incorporated an adversarial adaptation based on symmetric mapping to reduce the domain shift in EEG signals and the attention mechanism to enable cross-domain learning to capture both spatial-temporal relationships among the EEG electrodes. Rayatdoost et al. [11] proposed subject-invariant EEG-based representation learning, which uses the shared CNN model as a feature extractor and uses gradient inversion to minimize the loss of emotion recognition and maximize the loss of subject confusion. Bao et al. [31] proposed a two-level domain adaptation neural network, which used symmetric mapping to force the deep features closer to their corresponding class centers.

What these studies have in common is that they all learn a shared feature encoder by symmetric mapping and then eliminate the differences between different domains as much as possible through the gradient reversal layer for the discriminator. Unlike previous studies, our research attempts to combine the temporal sequence modeling ability of TCNs and adversarial adaptive learning based on asymmetric mapping to learn domain invariant features to realize cross-domain emotion recognition, which was not considered in earlier studies of EEG-based cross-domain emotion recognition.

3. Methods

3.1. Problem formulation and framework

Assume a source domain D_S is composed of a labeled EEG dataset X_S and a marginal probability distribution $P_S(X_S)$, while the target domain D_t is composed of an unlabeled EEG dataset X_t and a marginal probability distribution $P_t(X_t)$. The marginal probability distribution of the source and target domains is different, i.e., $P_t(X_t) \neq P_S(X_S)$. Domain adaptation solves the problem of different domain data distributions between the source domain and target domain.

The goal of our work is to regularize the learning of the source encoder M_S and target encoder M_t to minimize the distance between the

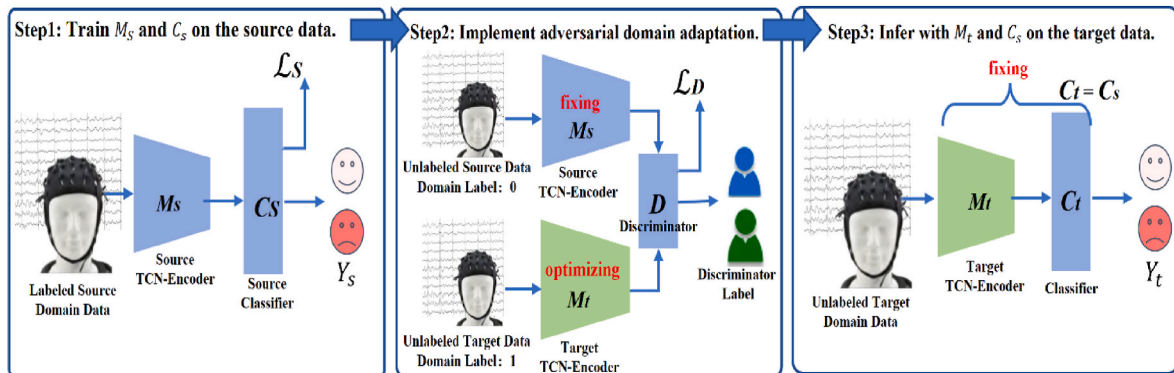


Fig. 2. Illustration of an AD-TCN framework in this work.

source mapping distributions $M_S(X_S)$ and target mapping distributions $M_t(X_t)$. This paper employs a framework consisting of three steps, as shown in Fig. 2. (1) Train source mapping M_S and source classifier C_S on the source data. M_S and C_S are pretrained using labeled sample data from the source domain. The source network model is equipped with characteristic information of emotion recognition through pretraining. (2) Implement adversarial domain adaptation. Then, the target encoder is adversarially trained so that the features extracted from the target encoder M_t have approximately the same distribution as those extracted from the source encoder M_S . (3) Infer with M_t and C_S on the target data. The classification performance is tested using the target encoder M_t and classifier C_S . For clarity, the algorithm is outlined with pseudocode (Algorithm 1).

Algorithm 1. Procedures of adversarial discriminative domain adaptation based on the TCN encoder for EEG-based emotion recognition.

Algorithm 1: Procedures of adversarial discriminative domain adaptation based on the TCN encoder for EEG-based emotion recognition.

Inputs: label EEG data from the source domain $D_S = \{X_S, Y_S\}$; unlabeled EEG data from the target domain $D_t = \{X_t, \cdot\}$;

Outputs: Trained target TCN-based encoder M_t ;
Predicted target domain labels Y_t ;

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1 Initialize  $\theta_S$ 
2 for  $X_S, Y_S$  in  $S_{(X,Y)}$  do
3   pretrain  $M_S$  and  $C_S$  based on loss below:
4    $\min_{M_S, C_S} \mathcal{L}_{cls}(X_S, Y_S)$ 
5    $= -\mathbb{E}_{(x_S, y_S) \sim (X_S, Y_S)} \sum_{k=1}^K \mathbb{I}_{[k=y_S]} \log C(M_S(x_S))$ 
6 end for
7 Set  $M_S = M_t, C_S = C_t$ 
8 repeat
9   for iteration-number do
10    updates  $\theta_D, \theta_t$  using Adam gradient descent
11     $\nabla_{\theta_D} (-\mathbb{E}_{x_S \sim X_S} [\log D(M_S(x_S))] - \mathbb{E}_{x_t \sim X_t} [\log(1 - D(M_t(x_t)))] )$ 
12     $\nabla_{\theta_t} (-\mathbb{E}_{x_t \sim X_t} [\log D(M_t(x_t))])$ 
13   end for
14 until convergence
15 return predicted target domain labels  $Y_t$ ; trained target TCN-based encoder  $M_t$ .
```

3.2. Feature encoder-TCN

Many studies [32,33] use the portability of deep neural networks to find the common space between different domains. However, neural networks such as convolutional neural networks (CNNs) have difficulty guaranteeing the representation invariance of feature graphs in different domains. Temporal convolutional networks (TCNs) were originally proposed by Bai et al. [34]. They demonstrated that TCNs are more efficient than recurrent neural networks (RNNs) for sequence modeling tasks in computer vision and natural language processing. Inspired by these findings, for EEG signals with specific temporal attributes, we use the temporal sequence model-TCN as the encoder for adversarial discriminative domain adaptation to realize cross-domain emotion recognition.

A TCN is a special CNN that aims at the convolution of 1D space. It can iteratively capture the long-term relationship of multiple layers. At present, TCNs are mainly used to deal with natural language processing, speech recognition, gesture recognition, etc. Its key is causal dilated convolution. Fig. 3(a) shows the illustration of casual dilated convolution with kernel size $k = 3$ and dilated factor $d = 1, 2, 4$.

A causal convolution is a convolution model for dealing with sequence problems, in which each node can consider the nodes before it. When the number of convolutional layers increases, there will often be negative effects such as gradient disappearance and overfitting. The TCN introduces a dilated convolution to solve these problems. A dilated convolution uses a dilated rate to express the size of the dilated layer, which can improve the receptive field. A 1D dilated convolution operation is defined as:

$$g(x) = \sum_{l=0}^{k-1} h(l)f(x-d \cdot l) \quad (1)$$

where f^* is the input, g^* is the output of the dilated convolution, $h(l)$ denotes the filter of length k , d is the dilation factor, and $x-d \cdot l$ represents the past direction of element x .

In this work, three residual blocks are used in the TCN, as shown in Fig. 3(b). Within the residual block, the TCN has two types of layers: a dilated causal convolutional layer and a ReLU layer. Weight normalization was applied to the convolutional filters for the purpose of normalization. Additionally, a dropout was added after each dilated convolution for regularization.

3.3. Adversarial adaptation by training a target encoder

A deep adversarial adaptation method based on symmetric mapping was originally proposed in Ref. [28] for image classification, which is

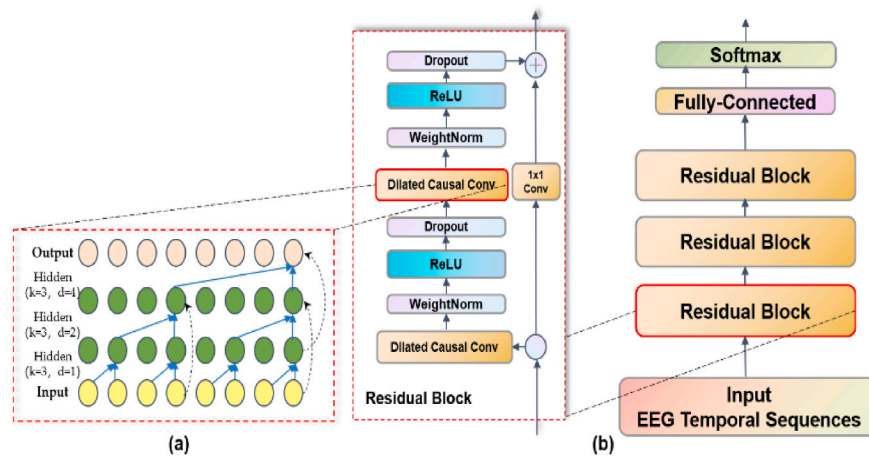


Fig. 3. The main architecture of the encoder-TCN. (a) Causal dilated convolutions with filter size $k = 3$ and dilated factors $d = 1, 2, 4$. (b) The total structure of feature encoder with three residual blocks.

used to solve cross-subject emotion recognition in Refs. [11,13,31]. Fig. 4 shows us the adversarial adaptation method based on symmetric mapping, which can be divided into three parts: a feature encoder, a discriminator D , and a classifier C .

When the network propagates forward, symmetric mapping $M(x)$ is obtained by using the shared feature encoder. As the source domain data are labeled, classification prediction loss $\mathcal{L}_C$ can be further calculated. The discriminator D distinguishes the domains of the features obtained after symmetric mapping and obtains the discriminator loss $\mathcal{L}_D$. In the process of back propagation, the gradient of the discriminator is calculated and transmitted to the gradient reversal layer (gradient multiplied by coefficient $-\lambda$), so the feature encoder will be updated in the direction of maximum discriminator loss. When the domain discriminator cannot distinguish the feature from the source domain or the target domain, the differences between domains have been eliminated as much as possible.

Unlike symmetric mapping, which uses a shared feature encoder to complete the feature mapping, asymmetric mapping allows independent source and target mappings by untying the weights and allowing more domain-specific feature extraction to be learned. In this way, the underfitting problem caused by too few parameters in symmetric mapping is avoided. In this work, we adopt asymmetric mapping to realize adversarial adaptation, which was not considered in earlier cross-domain EEG-based emotion recognition studies.

Specifically, the pretrained source model is used as an initialization for the target representation space and fixes the source model during adversarial training. As shown in Fig. 5, the target domain encoder M_t is initialized with the weight of the source domain encoder M_s . The reason why the weight of M_s is shared with M_t is that the domain discriminator D cannot distinguish which data distribution domain the EEG feature comes from at the beginning, thus avoiding the imbalanced training problem caused by the excessively strong discriminator in the early stage of adversarial learning.

After sharing the weights of M_s and M_t , the parameter weights of M_t will be revised in several rounds of adversarial learning. M_s and M_t extract features from the source domain data and target domain data, respectively, and then the domain discriminator D will distinguish which domain these encoded features come from. Furthermore, the labels of 0 or 1 are given to indicate that the feature information comes from the source domain or the target domain, respectively. Specifically, we begin by optimizing $\mathcal{L}_{cls}$ over M_s and C by training using the labeled source data, X_s and Y_s . The optimization equation is as follows:

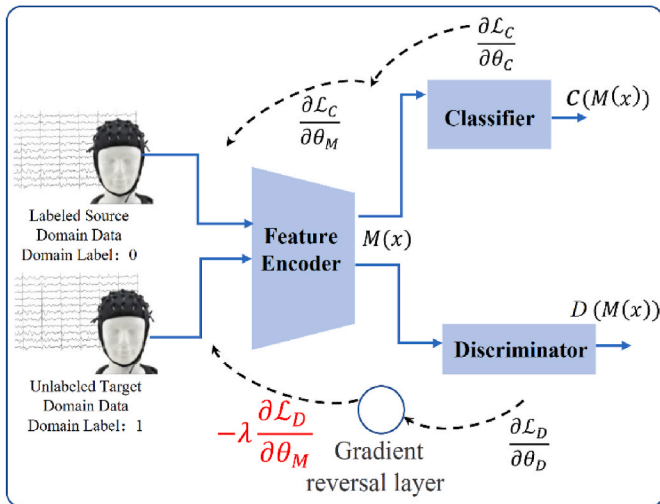


Fig. 4. An architecture deep adversarial domain adaptation based on symmetrical mapping.

$$\min_{M_s, C} \mathcal{L}_{cls}(X_s, Y_s) = -\mathbb{E}_{(x_s \sim y_s) \sim (X_s, Y_s)} \sum_{k=1}^K \mathbb{I}_{[k=y_s]} \log C(M_s(X_s)) \quad (2)$$

Because we opted to leave M_s fixed while learning M_t , we can optimize $\mathcal{L}_{adv_D}$ and $\mathcal{L}_{adv_M}$ without revisiting the first objective term. Corresponding to the following unconstrained optimization:

$$\min_D \mathcal{L}_{adv_D}(X_s, X_t, M_s, M_t) = -\mathbb{E}_{x_s \sim X_s} [\log D(M_s(x_s))] - \mathbb{E}_{x_t \sim X_t} [\log(1 - D(M_t(x_t)))] \quad (3)$$

$$\min_{M_t} \mathcal{L}_{adv_M}(X_s, X_t, D) = -\mathbb{E}_{x_t \sim X_t} [\log D(M_t(x_t))] \quad (4)$$

where D stands for the discriminator, which is mainly used for features extracted by the source domain encoder M_s and target domain encoder M_t . Through adversarial training, it is used to narrow the distribution of the features extracted by both the source domain and target domain. At the same time, the discriminator D is confused so that discriminator D cannot determine the origin of the data. This adversarial learning method makes the distribution of features extracted by M_t and M_s similar. In the training process, the discriminator D is optimized after fixing the parameters of M_t , and then the discriminator D is fixed to optimize M_t .

In the final stage, M_t and C are used to test the classification performance. M_t is used to capture the features in the target domain data, that is, mapping sample data in the target domain to the shared feature space. Then, the source domain classifier C_s obtained in the first stage is equivalent to C_t for classification, and the sample label y_t of the target domain is only used to evaluate the model performance.

3.4. Classifier

The classifier of our AD-TCN is composed of a fully connected layer with the activation function softmax. The outputs of the TCN-encoder are flattened and concatenated as a feature representation vector h . This feature representation vector h is fed into the fully connected layer, while a softmax function is used as an activation function. The function f is the full connection operator, which is a linear transformation:

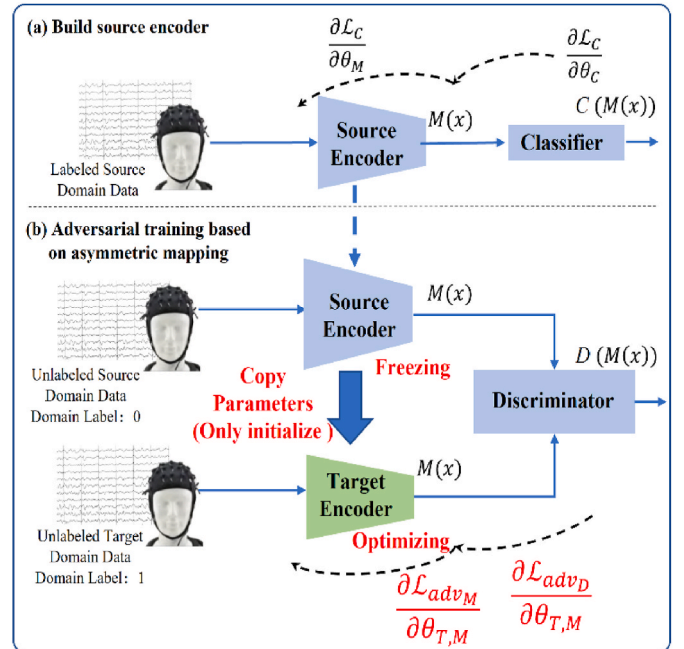


Fig. 5. An architecture deep adversarial domain adaptation based on asymmetric mapping. (a) build source encoder; (b) adversarial training.

$$f(h) = Wh + b = [y_1, \dots, y_c] \quad (5)$$

where W represents the transform matrix, b denotes the bias, and c is the number of emotional state categories.

The activation function softmax is primarily used for the task of classification. It can provide probability values for each emotional state category where the maximum value is returned as the final emotional state, which can be expressed as

$$P(j|X_n) = \frac{\exp(y_j)}{\sum_{i=1}^c \exp(y_i)}, j = 1, \dots, c \quad (6)$$

$$S = \operatorname{argmax} P(j|X_n) \quad (7)$$

where $P(j|X_n)$ represents the probability of a sample X_n belonging to the emotion state category j and S denotes the predicted human emotional state.

3.5. The implementation details

The infrastructure of our feature encoder-TCN is a neural network with three residual blocks. In the stage of training the source domain model, we use a batch size of 64 to train the source domain model for 200 epochs. In adversarial discrimination training, the adversarial discriminator consists of three additional fully connected layers: 2 hidden layers and 1 adversarial discriminator output. In addition to the output, these additional fully connected layers use the ReLU activation function. The target encoders are trained for 300 epochs, with a batch size of 64, and the Adam optimizer with a learning rate of 10^{-4} .

4. Materials

4.1. Emotional representative dimension

Because of the subjectivity and uncertainty of emotion, the labeling of emotion categories in emotion recognition is a special problem. The commonly used pattern in emotion classification is the 2D model of emotion, which maps emotion to valence and arousal dimensions.

The 2D emotional model proposed by Russell [35] is covered by valence and arousal scales in both high and low directions, representing all emotions in four quadrants. As shown in Fig. 6, the valence scale ranges from sad to pleased. The arousal scale ranges from sleepy to excited. Compared with discrete emotions (happiness, sadness, fear, anger, surprise, disgust), the 2D emotional model of valence and arousal can explain most emotion changes with a higher degree of fineness.

4.2. Dataset description

We used two publicly benchmark accessible datasets (DEAP [8], DREAMER [9]) and EEG dataset recordings from our lab (called self-recorded) for the experiment.

DEAP. In the DEAP dataset, 32 healthy participants (16 males and 16 females) participated in the experiment. Each participant was asked to watch 40 1-min music videos. Their EEG signals were recorded from 32 electrodes. The sampling rate of the original EEG signal is 512 Hz and then downsampled to 128 Hz. After watching each video, participants rated the self-assessment manikins (SAM) for arousal, valence, dominance and liking on a continuous scale of 1–9.

DREAMER. The DREAMER dataset consists of EEG data from 14 EEG electrodes of 23 subjects (14 males and 9 females). To build the dataset, 18 movie clips were used to stimulate different emotions. The duration of each film clip is between 65 and 393 s. The SAM was used to obtain a subjective assessment of arousal, valence and dominance on five-point scales.

Self-recorded dataset. Thirteen healthy volunteers (8 males and 5 females) were recruited to attend the EEG acquisition experiment.

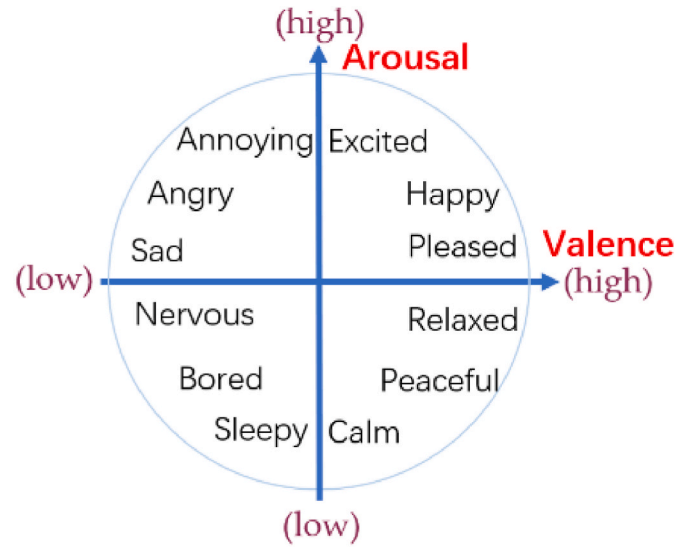


Fig. 6. Russell's arousal-valence 2D emotional representative dimension.

Written informed consent was obtained from all thirteen healthy subjects prior to the experiment. The rating scales were thoroughly explained. The recording session started only if the subjects expressed a sufficient understanding of the rating scales. Thirty-two electrodes were placed on the subject's head according to the international 10–20 system. The EEG signals are amplified and digitized at a sampling rate of 250 Hz.

Twenty video clips were used to evoke emotions based on the valence dimension during each trial, with 10 clips for each emotion category (i.e., sad and pleased). In the selection of videos, we first edited 140 comedy and tragedy videos (trials) from famous movies, each of which was approximately 30 s.

After each video (trial), an interface with the SAM scale shown in Fig. 7 is presented for 5 s to fill the valence score. The valence score is a measure of emotion from sadness to happiness (1–9). The GUI and paradigm are illustrated in Fig. 7.

4.3. Data processing

Data preprocessing. A bandpass filter between 1 Hz and 75 Hz is used to reduce noise and artifacts since raw EEG signals are often contaminated by electromyograph (EMG) signals from facial expressions and electrooculogram (EOG) signals from eye movements [19,36,37].

Feature extraction. Compared with other features (such as power spectral density (PSD), differential asymmetry (DASM), etc.), differential entropy (DE) features are more discriminative for emotion recognition according to previous studies [38,39]. We extract DE features in these four bands. Specifically, theta (4–7 Hz), alpha (8–13 Hz), beta (14–30 Hz), and gamma (31–50 Hz) were obtained by using short-term Fourier transform (STFT). DE features are extracted per second for these four frequency bands. DE can be calculated by following formula:

$$h(x) = - \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \log \left(\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right) dx \quad (8)$$

$$= \frac{1}{2} \log(2\pi e \sigma^2)$$

where σ is the variance of x and e is the Euler constant. Suppose an original EEG segment is represented as $S_k \in n \times t$, where n and t denote the number of electrode channels and the sample rate of raw EEG signals, respectively.

After the feature extraction stage, the original EEG segment is further

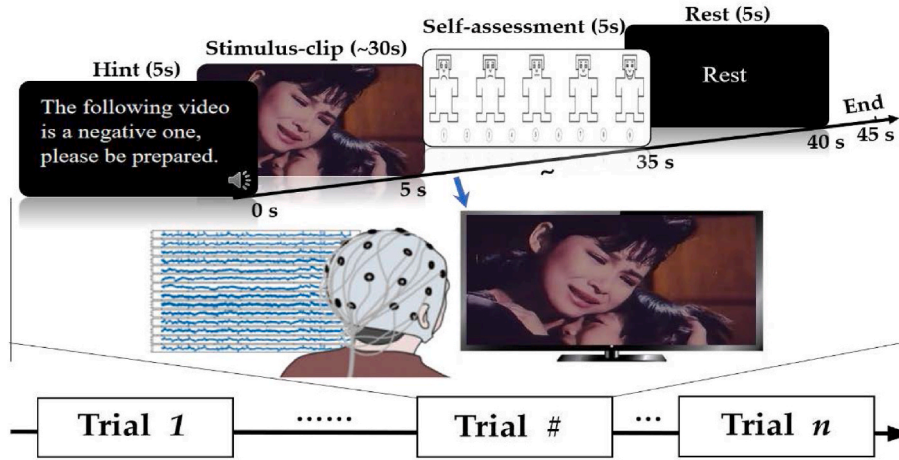


Fig. 7. Illustration of the data acquisition in this study.

transformed into the DE segment representation 1-dimensional vector $f \in \mathbb{R}^{n \times i}$, where i denotes the number of frequency bands. In this paper, i is set to 4 ($i \in \theta, \alpha, \beta, \gamma$).

Feature smoothing and scaling. In the field of EEG, emotions are usually assumed to be static to simplify the process of data processing, but we should consider continuous and subtle changes in the time domain. Previous studies [3,13,37] have shown that considering the dynamic changes of EEG during emotional stimulation can improve the performance of recognition. Therefore, the smoothing feature helps to filter out those components that have nothing to do with the emotional state. In this article, we use the moving average to remove the amount of variation in subtle changes in emotional state.

Feature scaling is a basic work of data processing. When the numerical attributes of datasets have very large proportional differences, it often leads to poor performance of machine learning algorithms, so we need to scale the data to a predefined range of values. In this article, the scaling method of mean normalization consists of rescaling the range of features to scale the range in [-1, 1]. The general formula is given as:

$$z = \frac{x - x_{\text{mean}}}{x_{\text{max}} - x_{\text{min}}} \quad (9)$$

where z is the normalized data, x is the original feature, and x_{max} , x_{mean} and x_{min} are the maximum, average and minimum values of the feature

data, respectively.

Temporal input construction. To obtain as much temporal dynamic information as possible from the EEG signals, we concatenate the extracted signals of each second along a certain time length to form a 2-dimensional temporal feature matrix $X_n \in \mathbb{R}^{f \times t}$ ($n=1,2,\dots,N$; $f=n \times i$) as depicted in Fig. 8, corresponding to the n -th sample, where N is the number of total samples and f and t are the 1-dimensional DE feature vector and length of time windows, respectively. Yang et al. [40] investigated the impact of different time windows on the performance, and they found that a window length of 9 s and an overlap of 8 s were the most suitable. Inspired by their findings, we choose windows of 9 s and an overlap of 8 s throughout the experiment.

Calibration of classification threshold. For many studies [8,41] on DEAP and DREAMER datasets, binary emotion categories are usually generated based on a fixed threshold, and rating data are directly discretized into low and high arousal (or valence) states.

In addition, there are some studies [37,42] considering that the boundary of emotion is often fuzzy and each subject has a different score scale, they segment by adding gaps, discarding the EEG data of arousal and valence between 4.8 and 5.2, to ensure that the EEG signal corresponds to a clear target emotion. The performance derived from such an approach is not convincing as a representation of the algorithm's effect on the entire dataset. Meanwhile, the self-assessment of emotional state

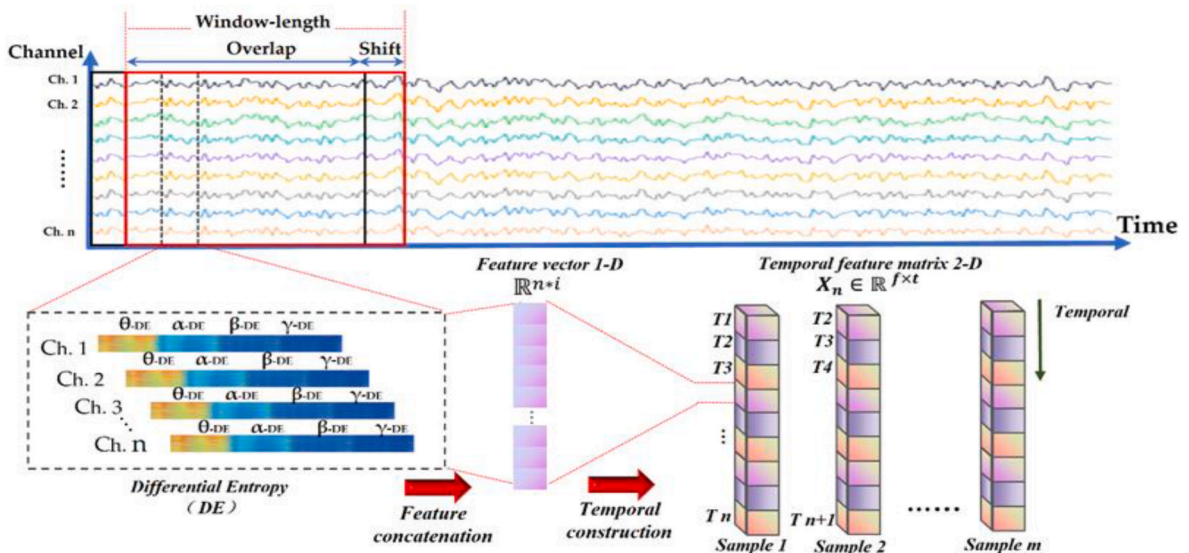


Fig. 8. The generation of temporal feature inputs.

is subjective [3]. A personal threshold generating dynamic emotional classes with self-assessed personalities could be more proper [43].

In this study, to generate appropriate emotional labels for the two categories, we determine the classification threshold by K-means clustering [44] on the subjective rating data of each subject, which is calculated by the midpoint of two clustering centers. It could avoid the malpractice of false high evaluation results caused by the extreme imbalance in the number of different emotional categories [40,43].

Fig. 9 shows an example of how to calibrate the threshold of personal emotion classification for subject 1 in the DEAP dataset. Fig. 9 shows that the two red dots are the clustering centers, which are [2.904286, 5.268571] and [7.564211, 6.092105]. The marker "⊗" represents that the midpoint of two clustering centers is [5.234248, 5.680338]. Low and high valence states could be defined by a threshold of 5.234248 (see Fig. 9(a)). Similarly, the value of 5.680338 is used as a threshold for dividing low and high arousal states (see Fig. 9(b)).

5. Experiments and results

In this section, to evaluate the effectiveness of AD-TCN in emotion recognition, we conducted two types of cross-domain experimental studies on the DEAP dataset, DREAMER dataset and self-recorded dataset: cross-subject, cross-dataset.

Note that EEG studies traditionally have some restrictions on experimental settings. The training dataset and testing dataset use the same experimental paradigm, the same equipment, and equipment with the same technical specification. To explore the effectiveness of our model under the setting of a great breakthrough in conventional EEG experiments (different subjects, different acquisition devices, different experimental paradigms, etc.), we validate our model on the cross-dataset domain adaptation task between datasets. These differences in the experimental setting may increase the existing differences between different subjects. It can be expected that cross-dataset adaptation is a more challenging task.

Regarding a performance evaluation, we report the average accuracy (ACC) and standard deviation (STD) in the DEAP dataset, DREAMER dataset, and self-recorded dataset, respectively. We calculate the prediction accuracy by using this equation:

$$\text{Accuracy} = \frac{|\{X : X \in \mathcal{D}_t \wedge \hat{y}(X) = y(X)\}|}{|\{X : X \in \mathcal{D}_t\}|} \quad (10)$$

where $\mathcal{D}_t$ is the set of test data, $y(x)$ is the truth label of x , and $\hat{y}(x)$ is the label predicted by our method.

5.1. Experiment I : Cross-subject transfer

To demonstrate the effectiveness of the proposed method, the leave-one-subject-out experiment for cross-subject emotion recognition was

evaluated on the DEAP DREAMER and self-recorded dataset. Specifically, one subject is used as the target data, while the source data are built with the rest of the subjects involved.

Since we use the unsupervised domain adaptation paradigm, the label of the target domain data is not available throughout the training process. SVM is the commonly used method to realize EEG-based emotion recognition [45,46]. Thus, we use linear SVM as a contrast method. To compare different types of classifiers, we use random forest (RF) [47] as the bagging classifier, XGBoost [48] as the boosting classifier, and hidden Markov model (HMM) [49] as the sequential classifier. In addition, we compared our proposed AD-TCN model with a variety of domain adaptation methods: TSVM, TCA, KPCA, and TPT.

Evaluation on DEAP dataset. For DEAP, the EEG data of the source domain are from 31 subjects, and the other subject is conducted as the target domain. The leave-one-subject-out experiment will be conducted in 32 runs. Fig. 10 (a) shows the accuracy of a total of 32 subjects. For valence classification, the average accuracy and standard deviation of all 32 subjects were 64.33% and 7.06%, respectively. For arousal classification, the average accuracy was 63.25%, and the standard deviation was 4.62%. Table 2 shows the experimental results of the proposed AD-TCN and other traditional domain adaptation methods on the DEAP dataset. Moreover, we perform a paired-sample *t*-test (measured in *p*-value) to examine whether the differences in recognition accuracies between AD-TCN and the other methods (SVM, RF, XGBoost, HMM, TSVM, TCA, KPCA, and TPT) are significant.

Among them, the performance of the classifier SVM is poor, and the average accuracy of valence and arousal is only 51.20% and 49.79%, respectively, which is close to random probability. The training sample may contain some information that has nothing to do with emotion, which may deviate from the SVM optimal hyperplane. It not only fails to achieve effective emotional recognition but also leads to negative transfer. For the other classifiers without domain adaptation (RF, XGBoost, HMM), accuracies of 50.35%, 57.70% and 49.53% were obtained in valence. For arousal, their accuracies were 53.49%, 53.72% and 52.26%, respectively.

Compared with the SVM, for the valence, the T-SVM, TPT, TCA, and KPCA improved by 10.57%, 6.23%, 5.03%, and 3.15%, respectively. Meanwhile, for arousal, the T-SVM, TPT, TCA, and KPCA outperformed the SVM by 6.8%, 4.97%, 2.02% and 8.36%, respectively. T-SVM learns decision boundaries in a semisupervised way, but some irrelevant source data may still be introduced in the training process. The TPT method mainly learns the mapping function by measuring the similarity between different data distribution pairs, but deviation is inevitable in the case of extremely dissimilar distributions. Two typical dimensionality reduction migration methods, TCA and KPCA, achieve effective knowledge transfer to a certain extent through feature reduction. However, the classification accuracy of the AD-TCN method is the highest, reaching 64.33% for valence and 63.25% for arousal, which is better than the four

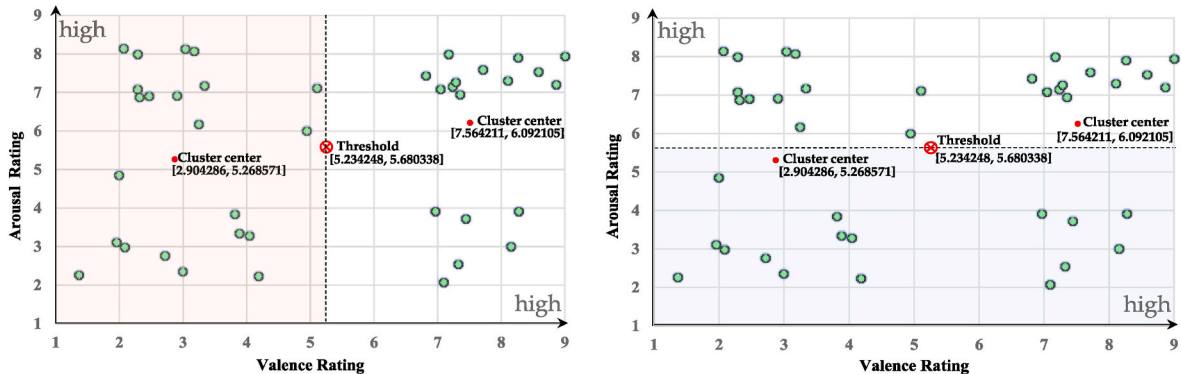


Fig. 9. Calibration of classification threshold for subject 1 in DEAP dataset. (a) determination of classification thresholds for valence dimension; (b) determination of classification thresholds for arousal dimension.

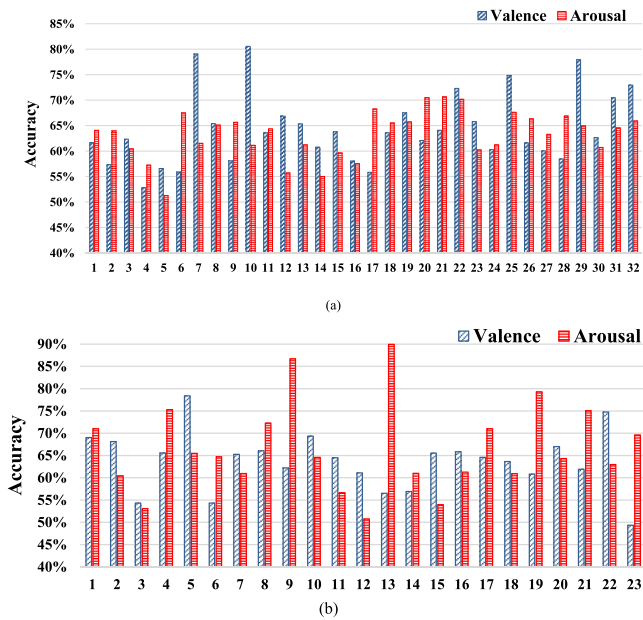


Fig. 10. Accuracy of various datasets. The X-axis represents the index of the target subject, and the Y-axis represents the accuracy. (a) Accuracy on the DEAP dataset; (b) Accuracy on the DREAMER dataset.

Table 2

Meaning accuracy (%) (and standard deviation) of different methods on the DREAMER and DEAP datasets. The differences in accuracies between AD-TCN and the other methods are highlighted with symbols (paired-sample *t*-test: ~nonsignificant, **p* < 0.05, ***p* < 0.01).

Method	DREAMER		DEAP	
	Valence	Arousal	Valence	Arousal
SVM	52.88** (10.46)	51.21** (14.64)	51.20** (10.97)	49.79** (16.27)
Bagging-RF [47]	54.57** (12.23)	52.94** (7.19)	50.35** (4.24)	53.49** (8.60)
Boosting-XGBoost [48]	57.83** (9.90)	58.32* (5.98)	57.70** (7.31)	53.72** (10.25)
Sequential classifiers-HMM [49]	48.22** (5.62)	50.34** (7.47)	49.53** (9.40)	52.26** (7.23)
T-SVM [23]	60.76* (9.77)	55.67** (12.07)	61.77* (8.93)	56.59** (11.98)
TPT [24]	59.22* (15.01)	61.89~ (13.18)	57.43* (14.54)	54.76 (12.48)
TCA [21]	55.85** (6.45)	54.37* (8.56)	56.23** (14.33)	51.81** (15.03)
KPCA [22]	53.74** (8.47)	60.03* (11.24)	54.35** (10.22)	58.15* (14.96)
Our AD-TCN	66.56 (10.04)	63.69 (6.57)	64.33 (7.06)	63.25 (4.62)

domain adaptation methods and significantly improved compared with the classifier SVM (paired-sample *t*-test, *p* < 0.05).

Evaluation on DREAMER dataset. The experimental setup for the DREAMER dataset is similar to DEAP. For the DREAMER dataset with 23 subjects, the leave-one-subject-out experiment will be conducted in 23 runs.

The recognition performance of the AD-TCN model on DREAMER is shown in Fig. 10 (b). The results are compared with the SVM, RF, XGBoost, HMM, T-SVM, TPT, TCA and KPCA algorithms, and the results are shown in Table 2. The accuracy of AD-TCN method in valence is 66.56%. Compared with SVM, RF, XGBoost, HMM, T-SVM, TPT, TCA and KPCA, the accuracy of AD-TCN is improved by 13.68%, 11.99%, 8.73%, 18.34%, 5.8%, 7.34%, 10.71% and 12.82%, respectively. The accuracy of AD-TCN in arousal is 63.69%, which is better than the

method without domain adaptation (SVM, RF, XGBoost, HMM) and the four domain adaptation methods (T-SVM, TPT, TCA, KPCA) of 12.48%, 10.75%, 5.37%, 13.35%, 8.02%, 1.8%, 9.32% and 3.67%, respectively.

Evaluation on self-recorded dataset. In addition, we designed the data acquisition paradigm to collect EEG data to further verify the effectiveness of our method. We use the self-recorded EEG dataset of 13 subjects in the valence dimension to conduct the cross-subject transfer experiment. Table 3 shows the accuracy of a total of 13 subjects for valence. On the self-recorded dataset, our method achieved stable and effective evaluation results, with an accuracy of 65.30% and standard deviation of 4.77% for valence.

5.2. Experiment II : Cross-dataset transfer

We investigate the performance of emotion classification for cross-datasets that are different across various experimental settings. As mentioned above, different emotional stimuli, subjects, and collection equipment were used on the DEAP, DREAMER, and self-recorded datasets. Therefore, this experiment can help us to study the effectiveness of the AD-TCN in the case of extreme data dissimilarity. By using data from different datasets as training data or test data in turn, we constructed eight classified task groups, which are as follows: (DEAP (valence) → DREAMER (valence), DEAP (arousal) → DREAMER (arousal), DREAMER (valence) → DEAP (valence), DREAMER (arousal) → DEAP (arousal), self-recorded (valence) → DREAMER (valence), DREAMER (valence) → self-recorded (valence), self-recorded (valence) → DEAP (valence), DEAP (valence) → self-recorded (valence)). The symbol A → B indicates that dataset A is the source data and dataset B is the test target data.

The domain adaptive method requires that the space of the source domain and the target domain are the same. The data acquisition equipment of the DREAMER dataset is a 14-channel EEG cap. To keep the feature space consistent, we only use 14 channels in the DEAP and self-recorded dataset corresponding to 6 task groups (DEAP (valence) → DREAMER (valence), DEAP (arousal) → DREAMER (arousal), DREAMER (valence) → DEAP (valence), DREAMER (arousal) → DEAP (arousal), self-recorded (valence) → DREAMER (valence), DREAMER (valence) → self-recorded (valence)). Note that the label dimension of our self-collected dataset was valence only. The classification accuracy of emotion recognition for the 8 cross-dataset task groups is shown in Table 4.

Note that our model is trained using the data in the DEAP dataset and tested on the DREAMER dataset. It produces a prediction accuracy of 62.36% for valence. Our model is trained using the data in the self-recorded dataset and tested on the DREAMER dataset. It yields a prediction accuracy of 62.65% for valence. These results are significantly higher than that of the classifier SVM (paired-sample *t*-test, *p* < 0.05). At the same time, the experimental results of other task groups are also higher than the accuracy of the classifier SVM, which is very promising in this case. This shows that our method can potentially reduce the domain shift between datasets.

6. Discussion

Our model provides a solution to overcome the problem of domain

Table 3

Accuracies of our method (AD-TCN) in the self-recorded dataset.

Target-Subject	Accuracy	Target-Subject	Accuracy
S 1	73.05%	S 8	71.94%
S 2	59.08%	S 9	63.38%
S 3	64.50%	S 10	60.96%
S 4	63.89%	S 11	65.96%
S 5	57.14%	S 12	64.58%
S 6	71.13%	S 13	66.67%
S 7	66.66%	Mean. (std.)	65.30% (4.77%)

Table 4

Averaged accuracies of our method (AD-TCN) in the cross-dataset.

Train	Test	Valence	Arousal
DEAP	DREAMER	62.36%	54.68%
DREAMER	DEAP	54.31%	55.88%
Self-recorded	DEAP	53.31%	–
DEAP	Self-recorded	55.47%	–
Self-recorded	DREAMER	62.65%	–
DREAMER	Self-recorded	54.42%	–

shift caused by cross-subject and cross-dataset and helps reduce the time required for labeled target domain collection data. The experimental results provide some inspiration for further research.

6.1. Cross-subject domain adaptation

We propose the AD-TCN method to study the cross-subject effectiveness in the leave-one-subject-out validation setting for domain adaptation. Each subject constitutes the target domain, while the rest of the subjects in the same dataset constitute the source domain.

We observed that the effect of our domain adaptation method on DEAP is not as effective as that on DREAMER and self-recorded datasets, which is consistent with the research [50]. There may be too many unrelated subsets among them and other factors that cause negative transfer and hinder the perfect application of domain adaptation.

Comparison with existing state-of-the-art studies. The focus of the comparison was on studies using the DEAP dataset for evaluation, as it is currently the most widely researched public dataset in the field of EEG emotion recognition and can be more easily validated externally.

We compare our proposed method with other different works for the DEAP dataset. Rayatdoost et al. [51] proposed a convolutional neural network fed by spectral topography maps from different bands for EEG-based emotion recognition and obtained the highest mean recognition accuracy of 59.22% for valence and 55.70% for arousal. Lew et al. [13] incorporated the attention mechanism to enable cross-domain learning to capture both spatial-temporal relationships among the EEG electrodes and an adversarial mechanism to reduce the domain shift in EEG signals. The average recognition accuracy rates were 56.60% for arousal and 56.78% for valence. Seeja et al. [52] exploited the image processing capability of CNN to classify different emotions from the scalogram images of EEG data. They reported model accuracy 61.50% for valence and 58.50% for arousal. Pandey et al. [20] proposed a subject-independent emotion recognition technique from EEG signals using variational mode decomposition (VMD) as a feature extraction technique and a deep neural network as the classifier. The highest average recognition accuracy rates were 61.25% for arousal and 62.50% for valence. Li et al. [53] presented their work on a deep belief network (DBN) to automatically extract high-level features from EEG. The highest average recognition accuracies on the DEAP dataset were 58.4% for arousal and 64.2% for valence. Miguel et al. [54] proposed a data transformation that seamlessly integrates individual traits into a cross-subject approach, improving classification results. The binary classification accuracies of arousal and valence are 59% and 64%, respectively. The results of relevant works are listed in Fig. 11. All these studies adopted leave-one-subject-out validation in the DEAP dataset. As shown in Fig. 11, our method achieved a higher average accuracy of 64.33% for valence and 63.25% for arousal on the DEAP dataset than previous studies.

Although the performance of some cross-subject studies based on the DEAP dataset is higher than ours, fair and accurate comparisons cannot be made due to the differences in paradigms of training and testing. The implementation of the transfer recursive feature elimination (T-RFE) algorithm proposed in Ref. [43] requires affective class tags from the validation set of the target domain. The reason is that SVM-based RFE feature sorting is naturally a supervised learning method. Wang et al.

[55] used labeled target data as calibration samples for model training.

In general, the performance of our proposed AD-TCN EEG-based emotion recognition is higher than that reported in most comparable studies [13,20,51–54]. Thus far, for the DEAP dataset, unsupervised domain adaptation cross-subject with EEG-based emotion estimation remains a greater challenge when there are unlabeled data in the target subject.

For the cross-subject experiment in the DREAMER dataset, we obtained average accuracies of 63.68% and 66.56% on arousal and valence, respectively. To a certain extent, the proposed method can effectively improve the emotion recognition performance of the cross-subject, surpassing the experimental performance of 62.17% and 62.49% by the dataset developer [9] on the subject dependence (not cross-subject). Since our self-recorded dataset has not been made public, we cannot give any benchmark results of other studies. The average accuracies and standard deviation of the self-recorded dataset are 65.30% and 4.77%, respectively, which proves that the performance of our method is promising.

Effectiveness analysis. To demonstrate the process of eliminating the domain difference of the feature distribution, we randomly selected the feature data of part of the test set and the corresponding feature data of part of the training set of a subject in the DEAP dataset and used T-SNE [56] to visualize the data distribution of the original target domain and source domain and the data distribution of the target domain and source domain after domain adaptation, as shown in Fig. 12.

In Fig. 12, the red font represents the source domain, and the blue font represents the target domain. The symbol "1" represents the characteristic data corresponding to high valence (positive), while "0" represents the characteristic data corresponding to low valence (negative). Fig. 12 (a) shows the original distribution of the source and target domains for the subject. As shown in Fig. 12 (a), EEG features have different distributions in the source domain and target domain, and the initial state mixes two types of emotions. Even when the source domain and the target domain are in the same emotional state, it is still very easy to confuse them, and the classification effect of general classifiers such as SVMs is poor. Fig. 12 (b) shows the feature distribution after domain adaptation by the AD-TCN method. Mapping features to the feature subspace effectively reduces the differences between the source domain and target domain, and the emotions of the same category between the source and target achieve alignment.

By comparing Fig. 12 (a) and (b), the EEG data of different subjects from the same emotion category are more obvious than the original DE feature aggregation, which is beneficial to the subsequent classification process. Therefore, for both source and target domains, the final state can well distinguish between the two types of emotions. There are still some instances that cannot be correctly identified. Individual differences limit the performance of EEG emotion recognition. However, most instances can be correctly predicted, which can explain the effectiveness of our AD-TCN.

6.2. Cross-dataset domain adaptation

In addition to the cross-subject domain adaptation, we evaluated the

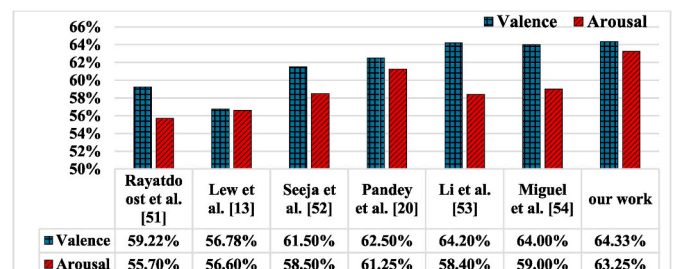


Fig. 11. Accuracy comparison between relevant works.

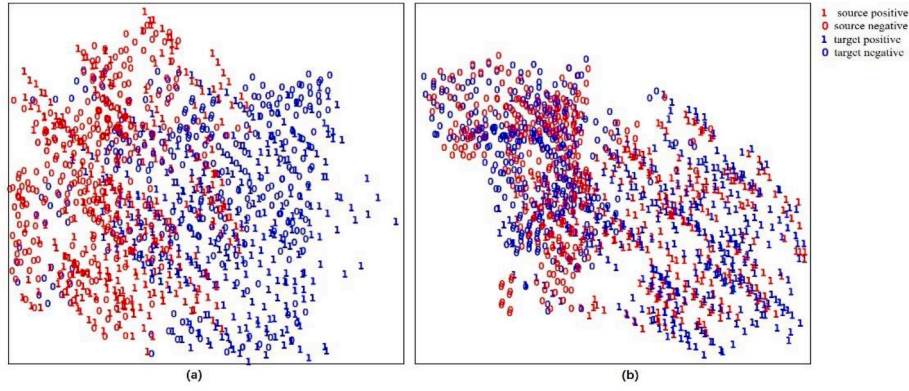


Fig. 12. Feature distributions of the two-dimensions. (a) Original feature distribution. (b) Feature transformed by our AD-TCN method.

performance of our method for emotion classification on cross-dataset experiments. The three datasets used in the experiments were different in various experimental settings, such as EEG acquisition devices and experimental paradigms.

In the 8 experimental cross-dataset task groups, we demonstrated the effectiveness of AD-TCN in challenging domain transfer cross-datasets through the demonstration of experimental results.

Compared with the SVM, although the accuracies of some task groups were significant (paired-sample *t*-test, $p < 0.05$), the accuracy was still lower than the cross-subject accuracy within the dataset. In addition to differences among subjects, differences in experimental settings (EEG acquisition devices, experimental paradigms, etc.) between the two datasets may lead to huge differences in sample distribution between the source domain and the target domain. Overall, cross-dataset attempts are of great practical interest, as they could help to overcome the limitations of traditional EEG-based applications that need to use the same experimental settings in the training and testing sessions.

6.3. Parameter sensitivity analysis

To investigate the change in the performance considering different hyperparameters, we performed an analysis of parameters sensitive to the DEAP dataset. Our AD-TCN is calculated for EEG signals consisting of temporal sequence data, implying that it is important to observe emotions from multiple window lengths and shifts, where overlapping equals window length minus shift.

We conducted an experiment while increasing the value of the window length from 3 to 12 s and changing the value of the shift between 1 s and 2 s to determine the appropriate window-length and shift. Note that when testing a specific window-length parameter, the shift parameter was fixed and vice versa. Based on the results in Table 5, when the window-length and shift are set to 9 s and 1 s, respectively, the method performs well. This observation is consistent with previous findings [19]. Thus, we choose window-length and shift of 9 s and 1 s throughout the experiment.

The batch size and epoch are other important parameters for the stage of adversarial adaptive training. The results are shown in Fig. 13 (a) and Fig. 13 (b), respectively. Fig. 13 (a) shows the variation of the classification accuracy with batch size $\in \{16, 32, 64, 128, 256, 528\}$ when the parameter epoch is fixed to 300. Fig. 13(b) shows the variation of the classification accuracy as epoch $\in \{100, 150, 200, 250, 300, 350, 400\}$ when the parameter batch size is fixed to 64. From Fig. 13, we can see that AD-TCN can achieve better performance when the batch size is 64 and the number of epochs is 300. Thus, for all experimental task groups of adversarial adaptive training, we set the batch size to 64 and the epoch to 300.

Table 5

Performance comparison of the proposed method with different window-lengths and shifts (overlapping = window-length – shift).

shift	window-length	Valence	Arousal
1 s	3 s	57.24%	58.78%
	6 s	61.64%	60.12%
	9 s	64.33%	63.25%
	12 s	62.87%	62.38%
2 s	3 s	54.47%	57.41%
	6 s	58.02%	62.74%
	9 s	60.43%	60.17%
	12 s	59.33%	60.04%

6.4. Ablation study

In this part, we conduct an ablation study on the DEAP experiment to validate the effectiveness of each module in our method. Our method is built upon three modules: adversarial discriminative training,

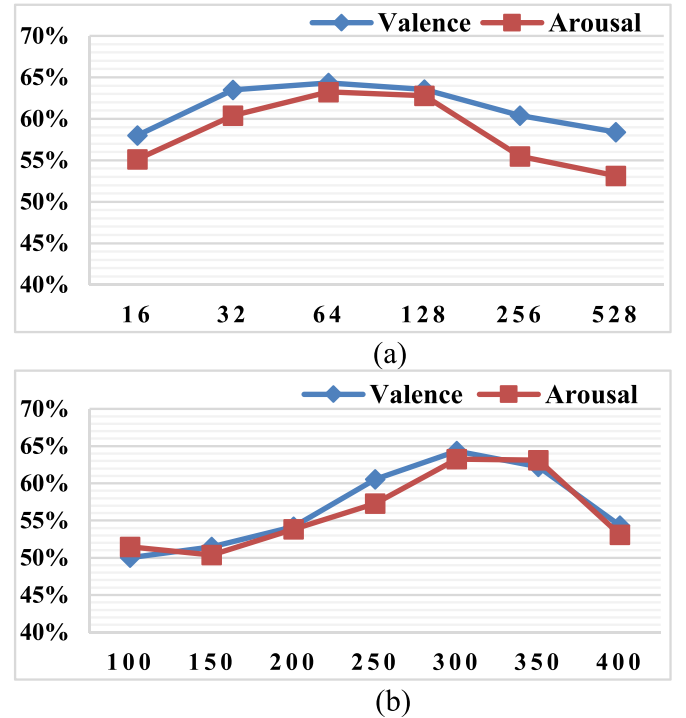


Fig. 13. Evaluation of AD-TCN with different batch sizes and epochs. (a) Average accuracy on the cross-subject task for different batch sizes; (b) Average accuracy on the cross-subject task for different epochs.

Table 6

The mean accuracies (%) (and standard deviation) of cross-subjects in an ablation study using different variant combinations on the DEAP dataset (paired-sample *t*-test, **p* < 0.05, ***p* < 0.01).

AD or not	Mapping	Feature encoder	Valence	Arousal
AD	Non-AD -	TCN	58.24* (8.23)	53.03** (6.25)
	symmetric	TCN	59.42* (5.92)	58.52* (5.01)
	asymmetric	MLP	56.75* (10.35)	57.18* (7.21)
		CNN	58.47* (8.74)	56.84* (7.80)
		RNN	60.55* (7.68)	57.52* (3.89)
		TCN	64.33 (7.06)	63.25 (4.62)

asymmetric-based mapping, and TCN-based feature encoder. Concretely, we compare our method with three variant combinations:

- Only the basic TCN model without adversarial adaptative training.
- The TCN as a feature encoder with adversarial adaptative based on symmetric mapping.
- Other classical deep neural networks (MLP (multilayer perception), RNN, and CNN) as feature encoders with adversarial adaptative based on asymmetric mapping.

The results of the ablation study are summarized in Table 6. Table 6 also gives the paired-sample *t*-test of our method and other comparison modules. The results demonstrate that our method shows the best performance overall. It is worth noting that the performance improvement of our proposed method is statistically significant compared to the corresponding model. Based on these results, we believe that the choice of TCN as an encoder and adversarial adaptative training based on asymmetric mapping is beneficial to improve performance.

6.5. Limitations and future research

The results show that our method is promising and superior to most of the reference techniques. The reason may be that joint temporal information and the adversarial discriminative domain adaptation method guarantee the representation invariance of feature graphs in different domains and fill in the differences between the different domains, thus effectively improving the performance of cross-domain emotion recognition.

However, at present, AD-TCN cannot completely eliminate the distribution differences, and the characteristic distribution of the source domain is still very scattered. Fig. 12 (b) shows the feature distribution map adapted by the AD-TCN domain. Some features of the source domain and the target domain are confused, and the features of the source domain and the target domain are relatively scattered and not clustered together. In the future, we will try to reduce the intraclass distance and expand the interclass spacing by adding the loss optimization strategy of interclass and intraclass spacing. In this way, the features of learning will be easier to distinguish than those currently learned.

6.6. Potential medical applications

Neurocognitive performance (NCP) indicates the psychological and cognitive ability of human beings to perform specific tasks [57]. The digitization and accurate evaluation of the NCP of patients with disorders of consciousness (DOC) is an open question.

In our previous work [58], it was proven that patients' cognitive levels can be reflected by their emotional performance. The proposed method is advantageous for practically developing an affective brain

computer interface for patients with DOC. It is very difficult to label for the EEG data of patients with DOC because patients with DOC suffer from severe motor impairments and generally cannot provide adequate emotional expressions [59]. Our method could reduce the reliance on the labeled data of the target patient. It could help to expand the application of EEG-based emotion recognition in patients with DOC.

7. Conclusion

It is an intractable and challenging task for cross-domain emotion recognition. Current studies focus on constructing subject-specific (same domain) models to recognize emotional states. Although the performance can be relatively high, these works need to collect a large amount of labeled data from the target domain before the training phase. In this paper, we combined temporal information and an adversarial discriminative domain adaptation method to evaluate the performance of cross-domain EEG-based emotion recognition. By taking advantage of encoder-TCN and adversarial discriminative training based on asymmetric mapping, cross-subject and cross-dataset validation experiments are conducted. The experimental results demonstrate that the framework of our method is an effective attempt at cross-domain emotion recognition without target domain labels during the training of the model. The method reduces the reliance on new subject labels and only requires unlabeled data to complete the establishment of new subject models, thus facilitating the fast acquisition of an emotion recognition model for new subjects.

Declaration of competing interest

All authors have read this manuscript and we declare that have no conflicts of interest to this work.

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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